Lecture 11: Hypersurfaces and the covariant derivative

An ***embedded regular hypersurface*** in $\mathbb{R}^{n+1}$ is a subset M which, near each point $p \in M$, is the regular zero set of one smooth real-valued function. There are an open neighborhood V of p and a smooth $F : V \to \mathbb{R}$ such that

$$M \cap V = F^{-1}(0), \qquad dF_q \neq 0 \quad (q \in M \cap V).$$

Here $n \geq 1$. The implicit function theorem says that M has dimension n: locally, one coordinate is a smooth function of the other n coordinates.

The tangent space and a local unit normal are

$$T_pM = \ker dF_p = \nabla F(p)^{\perp}, \qquad N(p) = \frac{\nabla F(p)}{\|\nabla F(p)\|}.$$

A hypersurface has codimension one. The familiar surfaces in $\mathbb{R}^3$ are the case $n = 2$.

Following Kühnel, a ***regular hypersurface element*** is a smooth map

$$f : U \longrightarrow \mathbb{R}^{n+1}, \qquad U \subset \mathbb{R}^n \text{ open},$$

whose differential has rank n everywhere (so f is an immersion). Write

$$u = (u^1, \dots, u^n), \qquad f_i = \frac{\partial f}{\partial u^i}, \qquad f_{ij} = \frac{\partial^2 f}{\partial u^i \partial u^j}.$$

The vectors $f_1, \dots, f_n$ are linearly independent, and

$$T_{f(u)}f = \operatorname{im} Df_u = \operatorname{span}\{f_1(u), \dots, f_n(u)\}.$$

The notation f_i means a derivative of the whole vector-valued map, not its i-th component.

After restricting U near a point, f parametrizes an embedded regular hypersurface. Choose n ambient coordinates whose derivative minor is invertible. The inverse function theorem makes these coordinates local parameters, and the remaining coordinate is a graph.

An immersion need not be globally injective. All arguments below are local, on one such parametrized patch.

The basic examples work in every dimension.

An affine hyperplane is

$$M = \{x \in \mathbb{R}^{n+1} : \langle x, a\rangle = b\}, \qquad a \neq 0.$$

Its normal $a/\|a\|$ is constant and $T_pM = a^\perp$ for every $p \in M$.

The sphere of radius $R > 0$ is

$$S^n_R = \{x \in \mathbb{R}^{n+1} : \|x\|^2 = R^2\}.$$

Taking $F(x) = \|x\|^2 - R^2$ gives

$$T_pS^n_R = p^\perp, \qquad N(p) = p/R.$$

For a smooth function $H : U \subset \mathbb{R}^n \to \mathbb{R}$, its graph has parametrization and unit normal

$$f(u) = (u, H(u)), \qquad \nu(u) = \frac{(-\nabla H(u), 1)}{\sqrt{1 + \|\nabla H(u)\|^2}}.$$

The first n rows of Df form the identity matrix, so this parametrization is regular.

The fundamental forms extend without changing their meaning. Choose a smooth local unit normal $\nu : U \to S^n$, and set

$$g_{ij} = \langle f_i, f_j \rangle, \qquad h_{ij} = \langle f_{ij}, \nu \rangle.$$

The matrix $g = (g_{ij})$ is positive definite. It represents the first fundamental form $I(X, Y) = \langle X, Y \rangle$ in the tangent basis $(f_1, \ldots, f_n)$.

The shape operator and second fundamental form are

$$L = -D\nu \circ (Df)^{-1}, \qquad II(X, Y) = I(LX, Y).$$

In particular, $L(f_i) = -\nu_i$ and $II(f_i, f_j) = h_{ij}$. Since $f_{ij} = f_{ji}$, the form II is symmetric.

For tangent vectors with coordinate columns $a, b \in \mathbb{R}^n$,

$$I(X, Y) = a^{\mathsf{T}} g b, \qquad II(X, Y) = a^{\mathsf{T}} h b.$$

Our goal is to differentiate tangent vector fields using only I. The construction will initially use the ambient space; Theorem 4.6 will show that the result is intrinsic.

A ***vector field along*** f is a smooth map $Y : U \to \mathbb{R}^{n+1}$, with $Y(u)$ regarded as a vector based at $f(u)$. It is ***tangent*** if

$$Y(u) \in \operatorname{im} Df_u \quad \text{for every } u \in U.$$

Every tangent field has a unique expression

$$Y = \sum_{j=1}^{n} \eta^j f_j,$$

where the $\eta^j : U \to \mathbb{R}$ are smooth coefficient functions. Both the coefficients and the basis vectors can vary with u.

For $X = \sum_i \xi^i f_i$ and a smooth scalar function $\varphi : U \to \mathbb{R}$, define

$$X(\varphi) = \sum_{i=1}^{n} \xi^i \partial_i \varphi, \qquad \partial_i = \frac{\partial}{\partial u^i}.$$

This is the rate of change of φ in direction X. In particular, $f_i(\varphi) = \partial_i \varphi$.

Superscripts on ξ^i, η^j, u^i are indices, not powers. We write all summation signs explicitly. All maps and fields in this lecture are smooth.

For a vector field Y on an open subset of $\mathbb{R}^{n+1}$, the ***directional derivative*** at p, in direction $X \in \mathbb{R}^{n+1}$, is

$$(D_X Y)_p = DY_p(X) = \frac{d}{dt} Y(p + tX)\bigg|_{t=0}.$$

Here DY_p is the ordinary derivative, represented by the Jacobian matrix of Y.

One can replace the straight line by any smooth curve $c(t)$ with $c(0) = p$ and $\dot{c}(0) = X$. The chain rule gives

$$\frac{d}{dt} Y(c(t))\bigg|_{t=0} = DY_p(\dot{c}(0)) = DY_p(X).$$

Thus only the initial velocity of the curve matters. Its acceleration and its later motion are irrelevant.

For fixed Y, this expression is linear in the direction X. It requires the variation of Y near p, not just the single vector $Y(p)$.

Now suppose Y is defined only along the patch f. For $X \in T_{f(u)}f$, there is a unique $a \in \mathbb{R}^n$ with $Df_u(a) = X$. Define

$$(D_X Y)_{f(u)} = DY_u(a) = DY_u\big((Df_u)^{-1}X\big).$$

The curve $c(t) = f(u + ta)$ has $\dot{c}(0) = X$, and

$$(D_X Y)_{f(u)} = \frac{d}{dt} Y(u + ta)\bigg|_{t=0}.$$

This definition does not require extending Y off the surface. If an ambient extension is used, the chain rule gives the same answer.

For $X = \sum_i \xi^i f_i$, the formula becomes

$$D_X Y = \sum_{i=1}^{n} \xi^i \partial_i Y.$$

In particular,

$$D_{f_i} Y = \partial_i Y, \qquad D_{f_i} f_j = f_{ij}.$$

Even when X, Y are tangent, $D_X Y$ need not be tangent.

The unit cylinder provides the simplest warning. Take

$$f(u, v) = (\cos u, \sin u, v), \qquad \nu = (\cos u, \sin u, 0).$$

The tangent field

$$Y = f_u = (-\sin u, \cos u, 0)$$

has unit length, but its derivative in its own direction is

$$D_{f_u} Y = f_{uu} = (-\cos u, -\sin u, 0) = -\nu.$$

The derivative is entirely normal to the cylinder.

Ambient differentiation sees the bending of the cylinder in $\mathbb{R}^3$. To measure change *within the tangent spaces*, we keep only the tangential component of this derivative.

In this example that component is zero. This does not say that Y is constant as a vector in $\mathbb{R}^3$; it says that its change has no tangential component.

For tangent fields X, Y, their ***covariant derivative*** is the tangential projection of the ambient derivative:

$$\nabla_X Y = (D_X Y)^\top = D_X Y - \langle D_X Y, \nu \rangle \nu.$$

Thus $\nabla_X Y$ is again tangent. At a point it depends only on the value of X, but on the local variation of Y.

The orthogonal projection

$$P^\top(V) = V - \langle V, \nu \rangle \nu$$

does not change if ν is replaced by $-\nu$. Consequently the covariant derivative does not require an orientation: local normal choices give the same answer on overlaps.

For a scalar function there is no projection to perform. We write simply

$$\nabla_X \varphi = D_X \varphi = X(\varphi).$$

The symbols D and ∇ distinguish ambient differentiation from tangential differentiation.

The discarded normal component is exactly the second fundamental form. Since Y is tangent,

$$\langle Y, \nu \rangle = 0.$$

Differentiate this identity in direction X:

$$\langle D_X Y, \nu \rangle + \langle Y, D_X \nu \rangle = 0.$$

Our sign convention for the shape operator is $D_X \nu = -LX$. Therefore

$$\langle D_X Y, \nu \rangle = \langle Y, LX \rangle = II(X, Y).$$

We obtain the decomposition

$$D_X Y = \nabla_X Y + II(X, Y)\nu.$$

The first term is tangential; the second is normal. The covariant derivative records tangential change, whereas II records bending in the ambient space.

For the unit cylinder, $II(f_u, f_u) = -1$, so the formula recovers $D_{f_u} f_u = 0 - \nu$.

The usual rules of differentiation pass to ∇. For smooth scalar functions φ, ψ and tangent fields,

$$\nabla_{\varphi X+\psi Z} Y = \varphi \nabla_X Y + \psi \nabla_Z Y,$$

$$\nabla_X (Y + Z) = \nabla_X Y + \nabla_X Z,$$

and

$$\nabla_X (\varphi Y) = X(\varphi) Y + \varphi \nabla_X Y.$$

For the last identity, first differentiate in the ambient space and then project:

$$D_X (\varphi Y) = X(\varphi) Y + \varphi D_X Y.$$

The vector Y is already tangent, so its projection is itself.

Notice the asymmetry. Multiplication by a function can be pulled out of the *first* argument. In the second argument, there is an extra derivative term. Thus ∇ is not bilinear over smooth functions in both arguments.

The covariant derivative is ***compatible with the metric***:

$$X\big(I(Y,Z)\big) = I(\nabla_X Y, Z) + I(Y, \nabla_X Z).$$

To see this, differentiate the Euclidean inner product:

$$X\langle Y, Z\rangle = \langle D_X Y, Z\rangle + \langle Y, D_X Z\rangle.$$

Use the tangential-normal decomposition in each term. Because Y and Z are tangent, the normal components contribute zero:

$$\langle D_X Y, Z\rangle = \langle \nabla_X Y, Z\rangle,$$

$$\langle Y, D_X Z\rangle = \langle Y, \nabla_X Z\rangle.$$

In particular,

$$X\big(I(Y,Y)\big) = 2I(\nabla_X Y, Y).$$

This identity will let us recover the derivative of the coordinate tangent basis from the metric coefficients.

The ***Jacobi bracket***, also called the ***Lie bracket***, of two tangent vector fields is

$$[X, Y] = D_X Y - D_Y X.$$

Although the two derivatives can have normal components, their difference is tangent. Indeed,

$$\langle D_X Y - D_Y X, v \rangle = II(X, Y) - II(Y, X) = 0.$$

Consequently,

$$[X, Y] = \nabla_X Y - \nabla_Y X.$$

This is the ***torsion-free*** identity. It does not say that $\nabla_X Y = \nabla_Y X$ for arbitrary fields: their difference is the bracket.

In particular, coordinate tangent fields commute:

$$[f_i, f_j] = f_{ij} - f_{ji} = 0, \qquad \nabla_{f_i} f_j = \nabla_{f_j} f_i.$$

Here torsion-freeness is a property of the covariant derivative, not the torsion of a space curve.

The bracket can be computed without a normal vector or a metric. Write

$$X = \sum_{i=1}^{n} \xi^i f_i, \qquad Y = \sum_{j=1}^{n} \eta^j f_j.$$

The product rule gives

$$D_X Y = \sum_{i,j=1}^{n} \xi^i (\partial_i \eta^j) f_j + \sum_{i,j=1}^{n} \xi^i \eta^j f_{ij}.$$

In $D_X Y - D_Y X$, the second-derivative terms cancel because $f_{ij} = f_{ji}$. Thus

$$[X, Y] = \sum_{j=1}^{n} \left(\sum_{i=1}^{n} (\xi^i \partial_i \eta^j - \eta^i \partial_i \xi^j) \right) f_j.$$

Equivalently, its j-th coefficient is

$$[X, Y]^j = X(\eta^j) - Y(\xi^j).$$

Coordinate fields commute, but fields with variable coefficients need not commute.

On the plane, let $e_x = (1, 0)$, $e_y = (0, 1)$, and choose

$$X = e_x, \qquad Y = x\, e_y.$$

Then

$$D_X Y = \partial_x (x e_y) = e_y, \qquad D_Y X = 0,$$

so

$$[X, Y] = e_y \neq 0.$$

Moving in the x-direction changes the speed prescribed by the field Y. Moving in the y-direction does not change the constant field X.

On the half-plane $x > 0$, these two fields are linearly independent, but they cannot both be coordinate basis fields for one coordinate system: coordinate basis fields must have zero bracket.

On a Euclidean plane, $D = \nabla$. This example also shows directly why covariant derivatives do not commute when the two vector fields are interchanged.

A vector field acts on scalar functions as a first-order differential operator. The bracket is the commutator of these operators:

$$[X, Y](\varphi) = X(Y(\varphi)) - Y(X(\varphi)).$$

To verify this, expand both sides in coordinates. The terms involving $\partial_i \partial_j \varphi$ cancel, leaving the coefficient formula from the preceding pages.

This characterization is intrinsic: it does not mention the ambient Euclidean space or the first fundamental form.

The bracket is skew-symmetric and satisfies the ***Jacobi identity***:

$$[X, Y] = -[Y, X],$$

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

For a proof of the second identity, apply the left side to an arbitrary function. Expand the three commutators; every composition of three differential operators appears twice with opposite signs. A vector field is determined by its action on functions (in particular, on the coordinates).

Theorem (Kühnel 4.6). The covariant derivative is determined by the first fundamental form. In particular, it is intrinsic, even though we defined it using an ambient derivative and a projection.

Fix coordinates and write $g_{ij} = I(f_i, f_j)$. Introduce the scalar coefficients

$$\Gamma_{ij,k} = I(\nabla_{f_i} f_j, f_k).$$

Since the coordinate fields commute, torsion-freeness gives

$$\Gamma_{ij,k} = \Gamma_{ji,k}.$$

To prove the theorem, it is enough to recover these coefficients from g. Indeed, the inner products of a tangent vector with every basis vector determine that vector uniquely.

Then the product rule determines $\nabla_X Y$ for arbitrary fields X, Y from the derivatives $\nabla_{f_i} f_j$. The remaining step is to compute the $\Gamma_{ij,k}$.

Metric compatibility supplies three equations:

$$\partial_i g_{jk} = \Gamma_{ij,k} + \Gamma_{ik,j},$$

$$\partial_j g_{ik} = \Gamma_{ji,k} + \Gamma_{jk,i},$$

$$\partial_k g_{ij} = \Gamma_{ki,j} + \Gamma_{kj,i}.$$

For example, the first follows by differentiating $g_{jk} = I(f_j, f_k)$ in direction f_i.

Add the first two equations and subtract the third. Using symmetry in the first two indices of Γ, the unwanted terms cancel:

$$2\Gamma_{ij,k} = \partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}.$$

Therefore

$$\Gamma_{ij,k} = \frac{1}{2}\big(\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}\big).$$

Only the metric and its first derivatives appear. The second fundamental form and the normal vector have disappeared.

These coefficients are called the Christoffel symbols of the first kind (the terminology is introduced in 4.7).

Let $(g^{k\ell})$ be the inverse matrix of $(g_{k\ell})$, and write

$$\nabla_{f_i} f_j = \sum_{k=1}^{n} \Gamma_{ij}^{k} f_k.$$

Taking the inner product with f_ℓ gives $\Gamma_{ij,\ell} = \sum_k \Gamma_{ij}^{k} g_{k\ell}$, hence

$$\Gamma_{ij}^{k} = \sum_{\ell=1}^{n} g^{k\ell} \Gamma_{ij,\ell}.$$

Thus these coefficients, too, are determined by g. For $X = \sum_i \xi^i f_i$ and $Y = \sum_j \eta^j f_j$, the product rule now yields

$$\nabla_X Y = \sum_{k=1}^{n} \left(X(\eta^k) + \sum_{i,j=1}^{n} \xi^i \eta^j \Gamma_{ij}^{k} \right) f_k.$$

This proves Theorem 4.6. Knowing the metric as a function on a neighborhood, and the coefficient functions of X, Y, determines the covariant derivative.

The value of g at a single point is not enough: its first derivatives are needed. The conclusion concerns coordinate coefficients under the identification of tangent bases, not equality of ambient vectors on different hypersurfaces.

Compare the plane and the cylinder parametrizations

$$f(u,v) = (u,v,0), \qquad \widetilde{f}(u,v) = (\cos u, \sin u, v).$$

Restrict u to an interval of length less than 2π, so both are embedded patches. Both metric matrices are

$$(g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

All first derivatives of the metric vanish, so all $\Gamma_{ij,k}$ and Γ_{ij}^k vanish. On either patch, covariant differentiation simply differentiates the coefficients in the coordinate tangent basis.

Nevertheless, the ambient derivatives are different:

$$D_{f_u} f_u = 0, \qquad D_{\widetilde{f}_u} \widetilde{f}_u = (-\cos u, -\sin u, 0).$$

Their tangential components both vanish.

The correspondence $f(u,v) \mapsto \widetilde{f}(u,v)$ preserves the first fundamental form and the covariant derivative, but not the second fundamental form. This is the distinction between intrinsic and extrinsic geometry.